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CRTPostprocess for URP and Legacy Pipeline
by xionchannel software
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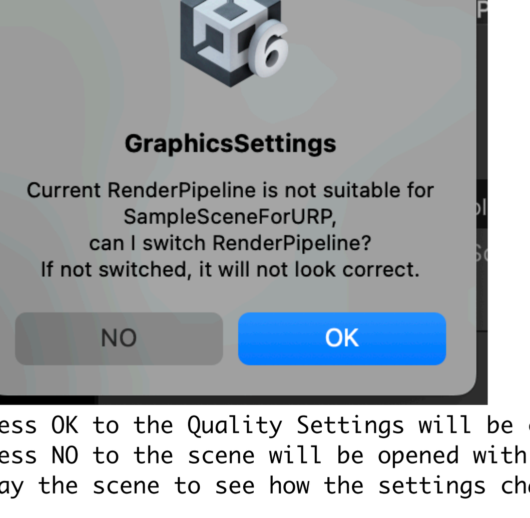
*** How to use sample scenes

Sample scenes are provided for each pipeline.
Please open the following scenes in an environment where the pipeline you are using is available.

/Assets/CRTPostprocess/SampleData/SampleScenes/

When a scene is opened, if the Editor environment does not match the scene settings,
a dialog box will appear asking if the settings should be changed automatically.



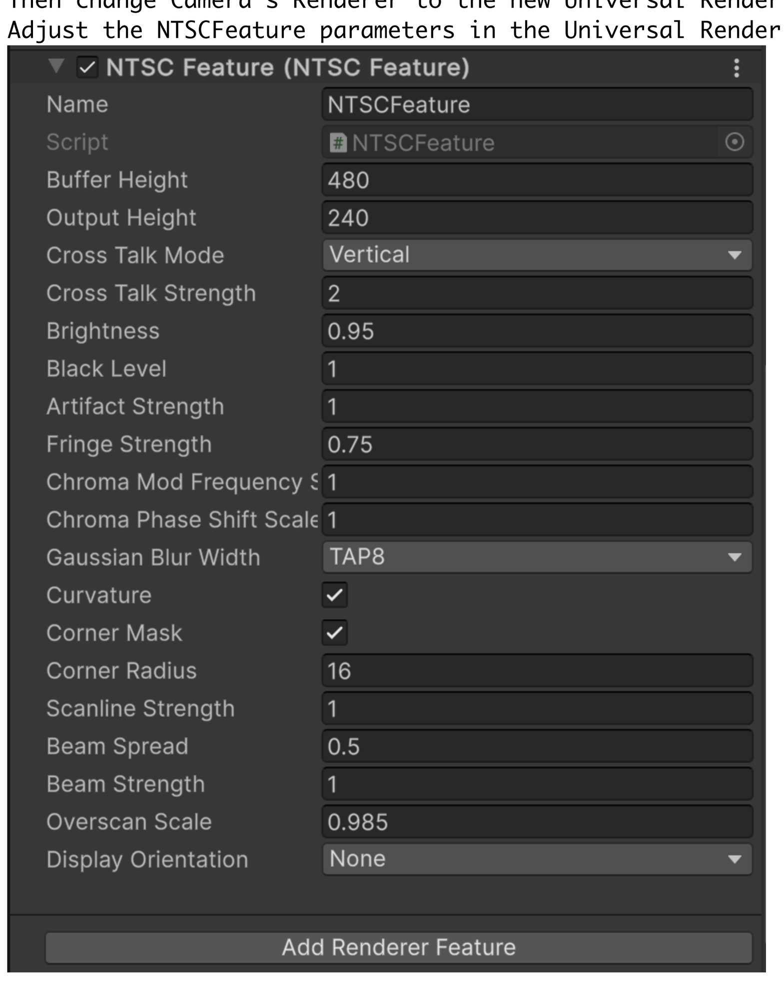
Press OK to the Quality Settings will be changed to the appropriate settings for the sample scene.
Press NO to the scene will be opened with incorrect setting that isn't changed.
Play the scene to see how the settings change with the program.

*** How to use postprocess

The setup procedure depends on the rendering pipeline.
See the procedure for your own rendering pipeline environment.

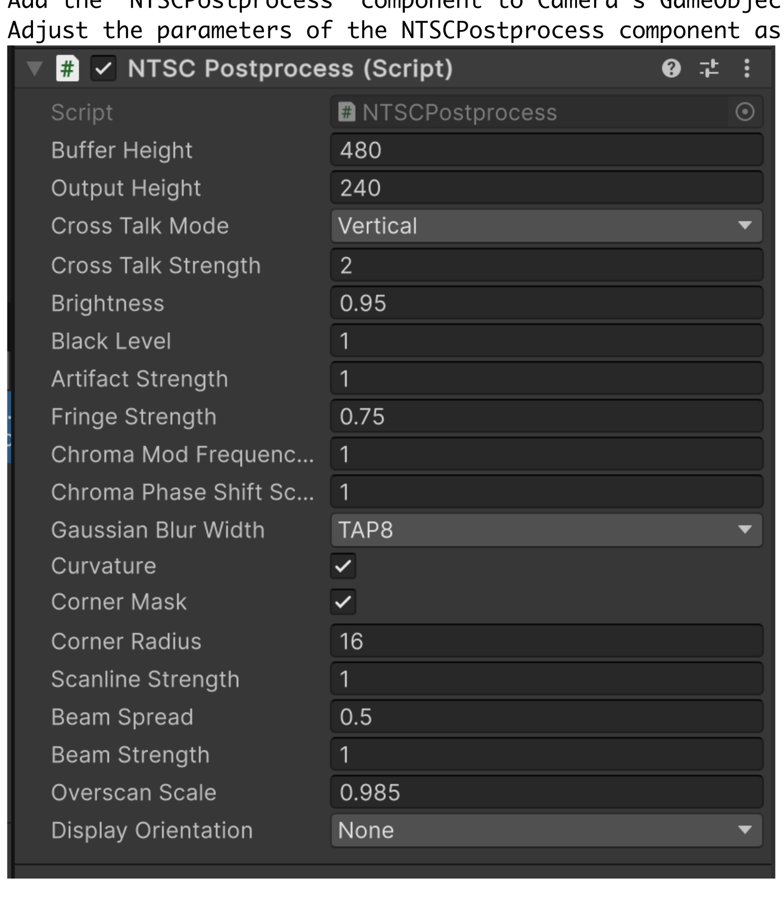
*** Setup in the URP Pipeline**

1. Add "NTSCFeature" to the current Universal Render Data. (e.g. "PC_Renderer")
   Alternatively, create a new Universal Render Data and add "NTSCFeature".
   (If you have not created a new one, proceed to the 4th step)
2. If you have created new Universal Render Data, add it to the current Universal Render Pipeline Asset.
3. Then change Camera's Renderer to the new Universal Render Data.
4. Adjust the NTSCFeature parameters in the Universal Render Data as needed.



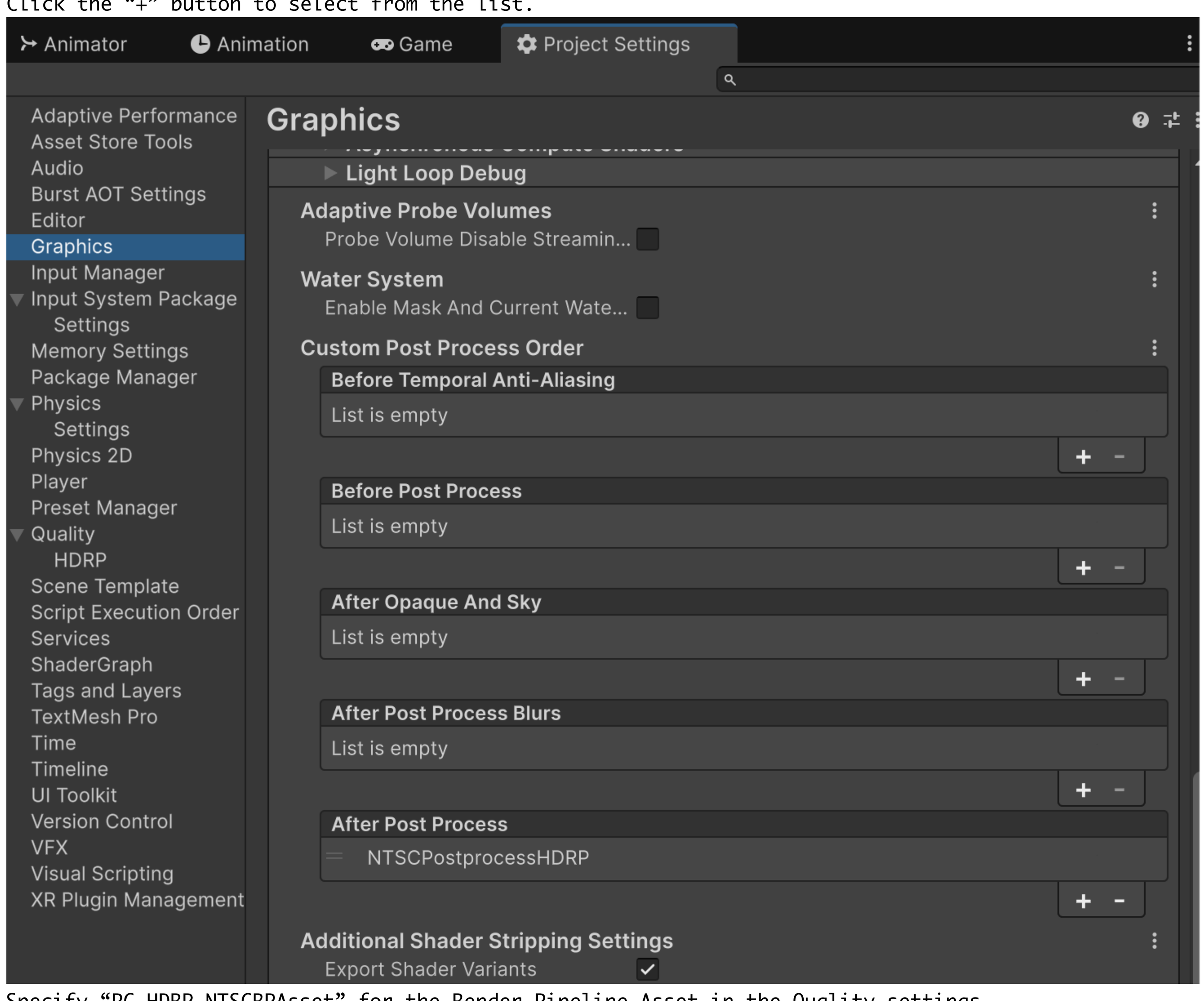
*** Setup in Legacy Pipeline (Built-in Pipeline)**

1. Add the "NTSCPostprocess" component to Camera's GameObject.
2. Adjust the parameters of the NTSCPostprocess component as needed.

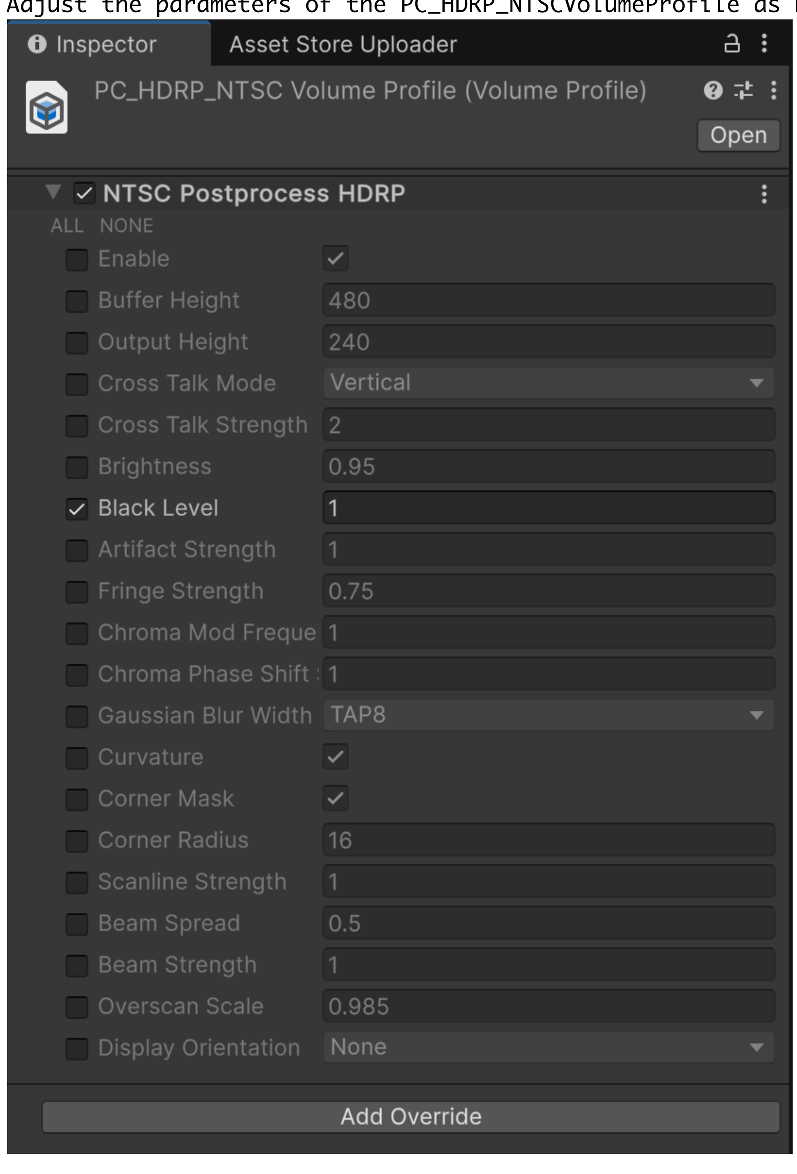


*** Setup in the HDRP Pipeline (Easy)**

1. Add "NTSCPostprocessHDRP" to After Post Process in the Custom Post Process Order in the Graphics setup.
   Click the "+" button to select from the list.



2. Specify "PC_HDRP_NTSCRPAsset" for the Render Pipeline Asset in the Quality settings.
   If you are creating a new Render Pipeline Asset, please see "Setup in the HDRP Pipeline (New)".
3. Adjust the parameters of the PC_HDRP_NTSCVolumeProfile as needed.

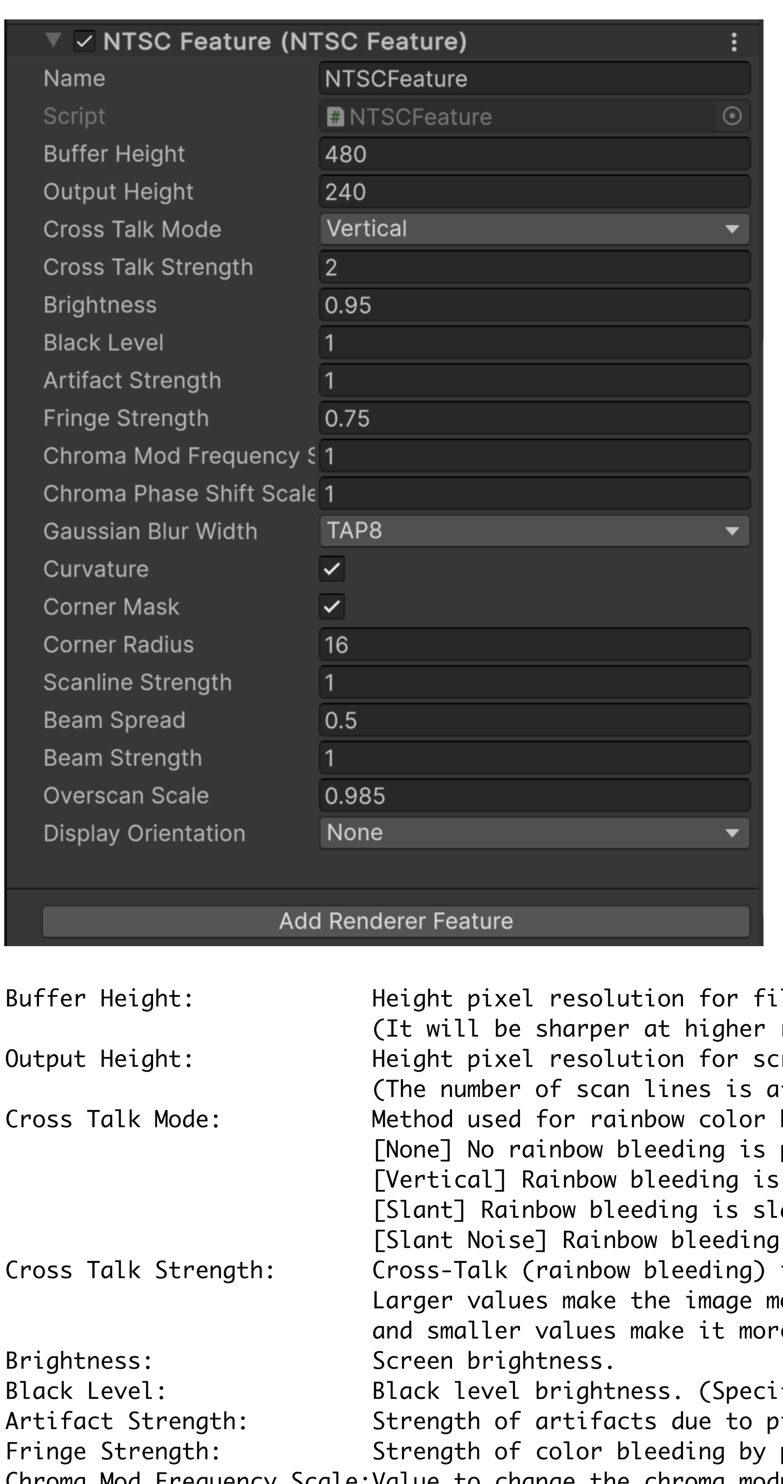


*** Setup in the HDRP Pipeline (New)**

1. Add "NTSCPostprocessHDRP" to After Post Process in the Custom Post Process Order in the Graphics setup.
   This is the same as "Setup in HDRP Pipeline (Easy)".
2. In the project view, create a "Volume Profile" asset from the "Create/Rendering/Volume Profile" menu.
3. Press the Add Override button from the Volume Profile's inspector
   and select "Post-processing/Custom/NTSCPostprocessHDRP" to create an override parameter for "NTSCPostprocessHDRP".
4. In the project view, create an "HDRP Asset" asset from the "Create/Rendering/HDRP Asset" menu.
5. Assign the volume profile asset to the Volume Profile field of the HDRP Asset just created.
6. Specify the HDRP Asset you just created for the Render Pipeline Asset in the Quality setting.
7. Adjust the parameters of the Volume Profile asset as needed.

*** Postprocess Parameters

The following are parameters for NTSCFeature, NTSCPostprocess, and NTSCPostprocessHDRP.
When using URP, use the parameters of NTSCFeature, and for Legacy (Built-in), use those of NTSCPostprocess.
For HDRP, use the parameters of the Volume Profile asset.

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Buffer Height:              Height pixel resolution for filter processing.
                            (It will be sharper at higher resolutions, but will lose its retro-display look.)
Output Height:              Height pixel resolution for screen output.
                            (The number of scan lines is affected by this height.)
Cross Talk Mode:            Method used for rainbow color bleeding using neighboring pixels.
                            [None] No rainbow bleeding is performed.
                            [Vertical] Rainbow bleeding is constant on the vertical axis.
                            [Slant] Rainbow bleeding is slanted.
                            [Slant Noise] Rainbow bleeding is slanted and flickering.
Cross Talk Strength:        Cross-Talk (rainbow bleeding) intensity.
                            Larger values make the image more vivid,
                            and smaller values make it more black and white.
Brightness:                 Screen brightness.
Black Level:                Black level brightness. (Specify 1.0526 for the same look as past versions.)
Artifact Strength:          Strength of artifacts due to pixels on the left and right.
Fringe Strength:            Strength of color bleeding by pixels on the left and right.
Chroma Mod Frequency Scale:Value to change the chroma modulation frequency. (default = 1)
Chroma Phase Shift Scale:   Value to change the chroma phase shift. (default = 1)
Gaussian Blur Width:        The blur width using neighboring pixels.
                            The larger the value, the more blurry the image.
Curvature:                  When turned on, the screen becomes distorted like a CRT.
Corner Mask:                Circular mask of screen corners; turning OFF eliminates the mask,
                            but off-screen pixels may be visible if Curvature is ON.
Corner Radius:              Number of pixel radius for circular masks in the corners of the screen.
                            (This value is used only when Curvature is OFF.)
Scanline Strength:          Strength of the scan line.
                            (0~1: the higher the value, the clearer the scan lines appear)
Beam Spread:                Scan line beam spread intensity. (0~1)
                            1: pixels appear to be connected to other scanlines above and below
                               if the pixel luminance is high.
                            0: pixels are not connected to other scanlines above and below.
Beam Strength:              The larger the value, the brighter the output pixels.
Overscan Scale:             Scale value of how much the drawing should be enlarged in relation to the screen.
                            A value less than 1 means the drawing will be enlarged;
                            a value greater than 1 means the drawing will be reduced.
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